Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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|---|---|---|---|---|
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## 2. Technology Overview

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|---|---|---|---|---|
| Title of Proposed Technology | *Post-Consumer Recyclates: collection, evaluation, and quality improvement via tailored additives* | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (x), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1)] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Chemicals* | | *Chemicals* | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Polymeric Materials* | *Polymer Recycling Technologies* | *Fine Chemicals* | *Additives* |
| Concept of Proposed Technology | *Core concept: Establish an analytics-driven, additivation-based framework to lift post-consumer recyclate (PCR) quality (PP/PE/PA/PET/ABS) to meet demanding application specs. Combine feedstock auditing (composition, contaminants, VOC/odor) with tailored additive packages (stabilizers, "repair" radical systems, chain extenders/couplers, compatibilizers, anti-odor sorbents).*<br>*Main functions:*<br>• *Quality mapping and batch equalization (reduce variability, contaminants, foreign polymers).*<br>• *Property repair and stabilization (processing and long-term stability, rheology, IV/MFR, color/odor/VOC), optional flame retardancy/reinforcement where needed.*<br>• *Process robustness.*<br>*Potential uses: Automotive (interior/exterior/structural), E&E housings and brackets, appliances/consumer goods, and building components subject to material-specific compliance.* | | | |

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| **Rationale for Joint R&D Project** | *Need: PCR streams show high batch-to-batch variability, aging-induced degradation, mixed polymers, contaminants, VOC/odor, and processing/tooling issues. To move PCR into high-value parts, robust analytics, targeted additive strategies, and pilot validation at industrial scale are required.*<br>*Collaboration rationale:*<br>• *Research institute: expertise in additive chemistry (stabilizers, chain extenders, rheology modifiers, hydrolysis stabilizers), structure–property–process analytics (FTIR/NMR/TGA/DSC, pyrolysis-GC/MS, VOC/fog/odor), recyclate rheology/repair, and test protocols; IP portfolio relevant to stabilization and odor control.*<br>• *Korean SMEs: access to large-scale collection/sorting, compounding and conversion lines, local testing/certification, and rapid path to domestic markets.*<br>• *Joint work bridges design-to-scale-up and accelerates qualification under realistic KR operating conditions and standards.* |
| **Key Words** | 1. *ProPCR upgrading via tailored additives*<br>2. *Variability reduction and rheology repair*<br>3. *Anti-odor/VOC mitigation*<br>4. *Compatibilization of mixed streams*<br>5. *Process/tool protection* |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level (5) | End of the Project | Level (7) |
|---|---|---|---|---|
| **R&D Objectives** | *Analytics and diagnostics toolkit: rapid contaminant screening (FTIR/NMR), polymer ID/blend ratio, MFR/IV tracking, black speck/fines metrics, VOC (VDA 278), odor (VDA 270), fog (DIN 75201), color ΔE, pyrolysis-GC/MS fingerprints.*<br>*Additive packages by polymer stream:*<br>• *PCR-PP/PE: processing and long-term stabilizers; rheology modifiers; coupling agents; compatibilizers for PE-in-PP (and vice versa); anti-odor sorbents; anti-acid packages.*<br>• *PCR-ABS/HIPS: AO/UV stabilization of PB phase, color/odor control, compatibilization for SAN/HIPS mixes, flame retardancy.*<br>• *PCR-PET: IV rebuilders (chain extenders), hydrolysis control, decontamination aid; color/yellow index control, fiber reinforcement for technical applications.*<br>• *PCR-PA: hydrolysis stabilizers and acid scavengers; processing AOs; chain repair (where applicable).*<br>*Optional functions: halogen-free FR where required; glass/minera/lcarbon fiber reinforcement; ESD/thermal conductivity modifiers (case-by-case).*<br>*LCA/ESG framework: CO2e per kg, recycled content, and processing energy hot-spots.* | | | |
| **R&D Target Specifications** | *Variability and stability:*<br>• *Batch-to-batch coefficient of variation (MFR/IV): −30% vs. baseline PCR within 12 months of operation.*<br>• *ΔMFR (or ΔIV PET) after 3× re-extrusions at defined thermal/shear history: ≤10%.*<br>*Contaminants/defects:*<br>• *Black specks/fish-eyes: −50% (count per kg or per m² sheet) vs. baseline.*<br>*Mechanical/appearance:*<br>• *Retain ≥90% of reference compound mechanicals (tensile/impact) at equal filler/fiber content; color ΔE* ≤1.0 for specified applications.*<br>*Application validation:*<br>• *Qualification of at least one automotive or E&E pilot part to OEM-relevant spec within project term.* | | | |
| **R&D Plan** | *Approach*<br>• *Build a data-rich view of incoming PCR streams → design polymer-specific additive modules → pilot on KR lines → validate with application parts and emissions/odor/mechanical standards → deliver guidelines and cost/LCA models.*<br>*Methodologies*<br>• *Analytics: FTIR/NMR, DSC/TGA, rheology (MFR/MVR/IV), microscopy/image analytics for specks, pyrolysis-GC/MS, VOC (VDA 278), odor (VDA 270), fog (DIN 75201), colorimetry.*<br>• *Processing: twin-screw compounding (masterbatch and direct), injection/extrusion trials, reprocessing loops.*<br>• *Validation: mechanical tests, aging (thermal/humidity/UV as relevant), emissions, and application-specific checks; LCA/ESG and cost modeling.*<br><br>*Key tasks*<br>• *T1 Feedstock mapping and rapid diagnostics pipeline; define KPIs per stream.*<br>• *T2 Design additive packages per polymer stream; screening DoE to rank efficacy vs. KPIs.*<br>• *T3 Lab-scale compounding on Fraunhofer-Lines; test specimen preparation; analytics*<br>• *T4 Multiple processing loops; demonstrating reprocessability*<br>• *T4 Pilot compounding and part manufacturing on KR lines; variability reduction loops.*<br>• *T5 Emissions/odor control.* | | | |

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- *T6 Application demo and OEM-style validation; standards compliance dossier.*
- *T7 LCA/ESG and scale-up economics; best-practice guideline for KR industry.*

*Milestones & timeline (3 years)*

| **Phase** | **Months** | **Milestone** | **Description** |
|---|---|---|---|
| I | 0–2 | M1 | Kick-off; stream selection; KPI/methods locked; risk plan |
| I | 0–6 | M2 | Feedstock analytics baseline; top-3 additive concepts per stream down-selected; preliminary lab scale compounding |
| II | 7–12 | M3 | Pilot compounding (KR) established; first property repair and variability reduction demonstrated |
| II | 13–18 | M4 | Emissions/odor packages validated (VOC/fog/odor targets hit on at least two streams) |
| II | 25–30 | M5 | Application demo parts meet OEM-style metrics; pre-certification data pack ready |
| II | 31–36 | M6 | Tech transfer; LCA/cost model; industry guideline and commercialization plan |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | *Technical contributions: additive concept design (stabilizers, radical "repair" systems, hydrolysis control, chain extenders/couplers, anti-odor); analytics (FTIR/NMR/DSC/TGA/py-GC/MS), VOC/fog/odor/color protocols; reprocessing stability; DoE and data modeling; LCA/ESG and cost/performance modeling.*<br>*Research scope: PP/PE/PA/PET/ABS PCR streams; compatibilization strategies for mixed streams; process/tool protection recommendations.*<br>*Resources: compounding and testing labs; emissions/odor analytics; project and data management; IP and tech transfer support.*<br>*Specific tasks: KPI definition; screening and pilot recipes; documentation for Phase II/main call; training of KR partners in methods.* |
| **Expected R&R of Korean SME Partner** | *Technical contributions: supply of PCR streams; sorting/cleaning and compounding scale-up; conversion (injection/extrusion/blow); pilot part fabrication; local standards/testing.*<br>*Development tasks: implement analytics pipeline; process window mapping; emissions/odor testing per KR/industry protocols; tool wear/fouling tracking and mitigation.*<br>*Testing/validation: coordinate mechanical/emissions/aging tests with KR test orgs; assemble data for Phase I/II gates; application-specific validation.*<br>*Resource commitments: pilot lines, materials/logistics, local regulatory liaison; cost/S&OP modeling; participate in tech transfer and commercialization.* |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | • *Technological achievements: proven additive toolboxes per major PCR streams; systematic variability reduction; robust VOC/odor mitigation; reprocessing-stable rheology; validated process/tool protection; application-grade parts.*<br>• *Commercialization potential: drop-in masterbatches/compound recipes; scale-up on KR lines; guideline and qualification data packages; joint IP where applicable.*<br>• *Industrial/economic impact: higher and more consistent PCR quality enabling entry into automotive/E&E; reduced scrap and downtime; lower total cost and CO2e per kg; strengthened KR–DE collaboration and export readiness.* |

## 6. References

(1) LBF stabilization and bio-based AO patents: WO2018153747; WO2022058371; WO2022090566; WO2020083740.

(2) Controlled hydrolysis and stabilization for condensation polymers (relevant to PCR-PET/PA): WO2021214207; WO2021213939.

(3) Anti-odor/low-emission part strategies: WO2022063723.

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
|---|---|---|
| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |

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